Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 21

Cardinality and Countability

Covers Chartrand & Zhang, Section 5.5

Status of this set

Ungraded practice. Solutions released with it. Shorter than usual, since Section 5.5 had one period rather than three.

Part I: How to Practice This Material

Everything reduces to one question

Is there a bijection? $|A| = |B|$ exactly when one exists. Denumerable means a bijection with $\mathbb{N}$ — equivalently, the elements can be *listed* with each appearing exactly once.
To show a set is denumerable: **produce a listing**. To show it is uncountable: argue that *no* listing can work, which usually means a diagonal argument.

Three mistakes to expect

1. **Assuming a proper subset must be smaller.** True for finite sets, false for infinite ones.
2. **Confusing "infinite" with "uncountable."** $\mathbb{Z}$ and $\mathbb{Q}$ are infinite and countable.
3. **Trying to disprove a listing by testing it.** You must rule out *all* listings.

Part II: Practice Problems

Section A — Countable or not?

1. Classify each as finite, denumerable, or uncountable, with a one-line reason.

(a) The set of even integers.

(b) $\{x \in \mathbb{R} : 0 < x < 1\}$.

(c) The set of integers divisible by 100.

(d) $\mathcal{P}(\mathbb{N})$.

Section B — Listings

2. Give an explicit listing showing that the set of odd integers is denumerable, and state the bijection it defines.

Section C — The diagonal argument

3. In Cantor's proof, the constructed number b has nth digit equal to 2 if $a_{nn} = 1$ and 1 otherwise.

(a) Given the list $a_1 = 0.5231\ldots$, $a_2 = 0.1104\ldots$, $a_3 = 0.9817\ldots$, find the first three digits of b.

(b) Why must b differ from every a_n?

(c) Why does the rule avoid the digits 0 and 9?

Section D — Writing

4. In five to eight complete sentences, explain why $\mathbb{Z}$ and $\mathbb{N}$ have the same cardinality even though $\mathbb{N}$ is a proper subset of $\mathbb{Z}$. Say why this cannot happen for finite sets, and connect your answer to a result from Week 7.

Section E — Challenge (optional)

5. Cantor's proof of $|A| < |\mathcal{P}(A)|$ builds $A_g = \{a \in A : a \notin g(a)\}$ and asks whether $a_0 \in A_g$ where $g(a_0) = A_g$. Explain why both answers lead to a contradiction, and say what this resembles from Lecture 1.